

Physical-Space Matrix Hypercontractivity over Large Alphabets for Target Exponents at Least Two

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Source Question

Arunachalam and Doriguello ask in Future work item 4 of arXiv:2109.02600 whether one can prove a general (q, p) -hypercontractive statement for matrices, first for matrix-valued functions over $\{0, 1\}^n$, and then for functions over $\mathbb{Z}_r^n$. The source passage is shown in Figure 1.

3. Better bounds on streaming algorithms. What is the quantum space complexity of approximating Unique Games? Given the recent work of Kallaugher and Parekh [45], who proved a $\Omega(n)$ streaming lower bound for Max- t -Cut, we conjecture that the same bound applies to Unique Games. A proof might require obtaining new quantum lower bounds for the communication problems introduced in [19, 20, 47, 48].

4. Generalized hypercontractivity. Another open question is regarding our main Result 1, which shows a form of $(2, p)$ -hypercontractivity, since the result works for all Schatten p -norms with $p \in [1, 2]$. Can we prove a general (q, p) -hypercontractive statement for matrices, firstly for matrix-valued functions over $\{0, 1\}$, and then further generalize that to functions over $\mathbb{Z}_r$? Proving this might also require a different generalization of the inequality of Ball, Carlen, and Lieb [5].

2 Preliminaries

Let $[n] \triangleq \{1, \dots, n\}$. For $r \in \mathbb{Z}$, $r \geq 2$, we let $\mathbb{Z}_r \triangleq \{0, \dots, r-1\}$ be the ring with addition and multiplication modulo r , and let $\omega_r \triangleq e^{2\pi i/r}$. Given $S \in \mathbb{Z}_r^n$, we write $|S| \triangleq |\{i \in [n] : S_i \neq 0\}|$ for its Hamming weight. Let $D(\mathbb{C}^m)$ be the set of all quantum states over $\mathbb{C}^m$, i.e., the set of positive semi-definite matrices with trace 1. For a matrix $M \in \mathbb{C}^{m \times m}$, its (unnormalized) Schatten p -norm is defined as $\|M\|_p \triangleq (\text{Tr } |M|^p)^{1/p} = (\sum_{i=1}^m \sigma_i(M)^p)^{1/p}$, where $\{\sigma_i(M)\}_i$ are the singular values of M , i.e., the eigenvalues of the positive semi-definite operator $|M| \triangleq \sqrt{M^\dagger M}$. Moreover, let $\|M\| \triangleq$

Figure 1: Future work item 4 in arXiv:2109.02600, PDF page 9.

This packet proves a physical-space theorem in the range $1 \leq p \leq 2 \leq q < \infty$. It is therefore stronger than the earlier coefficient-space packet for this target, but still does not settle the intermediate range $p < q < 2$.

Notation

Let $G = \mathbb{Z}_r^n$ with normalized counting measure and characters $\chi_S(x) = \omega_r^{S \cdot x}$, where $\omega_r = e^{2\pi i/r}$. For $f : G \rightarrow \mathbb{C}^{m \times m}$ write

$$\widehat{f}(S) = \mathbb{E}_{x \in G} f(x) \overline{\chi_S(x)}, \quad T_\rho f = \sum_{S \in G} \rho^{|S|} \widehat{f}(S) \chi_S.$$

Here $|S| = |\{j : S_j \neq 0\}|$. The mixed norm is

$$\|f\|_{L_s(S_s)} = (\mathbb{E}_{x \in G} \|f(x)\|_s^s)^{1/s},$$

where $\|\cdot\|_s$ is the unnormalized Schatten s -norm. For $1 \leq s \leq 2$ set

$$\zeta_r(s) = \frac{(s-1)(1-(s-1)^{r-1})}{(r-1)(2-s)}$$

with the continuous values $\zeta_r(1) = 0$ and $\zeta_r(2) = 1$. For $2 \leq q < \infty$ set

$$\eta_r(q) = \frac{(q-1)((q-1)^{r-1} - 1)}{(r-1)(q-2)}$$

with $\eta_r(2) = 1$.

Proof Intuition

The paper proves an r -point Ball–Carlen–Lieb convexity inequality: for $1 \leq p \leq 2$, the $L_p(S_p)$ norm of one $\mathbb{Z}_r$ coordinate controls a weighted Hilbertian sum of its matrix Fourier coefficients, with weight $\zeta_r(p)$ on all nonzero frequencies.

For $q \geq 2$ the Ball–Carlen–Lieb smoothness inequality gives the reverse estimate with constant $\eta_r(q)$. Applying this smoothness estimate to the noisy coefficients, using $\|A\|_q \leq \|A\|_p$, and then applying the source convexity estimate yields the one-coordinate $L_p(S_p) \rightarrow L_q(S_q)$ contraction whenever $\eta_r(q)\rho^2 \leq \zeta_r(p)$. The product result is obtained by treating all other coordinates as a finite direct sum of matrix algebras.

Auxiliary Inequalities

Lemma 1 (Source r -point convexity). *Let $1 \leq p \leq 2$ and let $A_0, \dots, A_{r-1}$ be matrices, or elements of a finite direct sum of matrix algebras. Then*

$$\left(\frac{1}{r} \sum_{x=0}^{r-1} \left\| \sum_{k=0}^{r-1} \omega_r^{kx} A_k \right\|_p^p \right)^{2/p} \geq \|A_0\|_p^2 + \zeta_r(p) \sum_{k=1}^{r-1} \|A_k\|_p^2.$$

Proof. For matrices this is exactly the r -point Ball–Carlen–Lieb inequality proved in Section 3.1 of arXiv:2109.02600. The same statement for finite direct sums follows by applying the matrix statement to the corresponding block-diagonal embedding; rescaling the trace only rescales both sides by the same homogeneous factor. $\square$

Lemma 2 (r -point smoothness). *Let $2 \leq q < \infty$ and let $A_0, \dots, A_{r-1}$ be matrices, or elements of a finite direct sum of matrix algebras. Then*

$$\left(\frac{1}{r} \sum_{x=0}^{r-1} \left\| \sum_{k=0}^{r-1} \omega_r^{kx} A_k \right\|_q^q \right)^{2/q} \leq \|A_0\|_q^2 + \eta_r(q) \sum_{k=1}^{r-1} \|A_k\|_q^2.$$

Proof. We mimic the block-diagonal proof of the source convexity inequality, using the Ball–Carlen–Lieb smoothness inequality in place of convexity. That smoothness inequality says that if the same trace-orthogonality condition used in the source proof holds for Z and W , then

$$\|Z + W\|_q^2 \leq \|Z\|_q^2 + (q - 1)\|W\|_q^2.$$

For $j = 0, \dots, r - 1$, define the r -block matrices

$$Z_j = \text{diag}(A_j, \omega_r^j A_j, \dots, \omega_r^{(r-1)j} A_j).$$

As in the source proof, the root-of-unity cancellation gives the required trace-orthogonality between Z_j and Z_k whenever $j \neq k$, and also between each selected Z_j and the remaining tail.

For any permutation $(k_1, \dots, k_{r-1})$ of $\{1, \dots, r - 1\}$, repeated application of smoothness gives

$$\left\| \sum_{j=0}^{r-1} Z_j \right\|_q^2 \leq \|Z_0\|_q^2 + \sum_{\ell=1}^{r-1} (q - 1)^\ell \|Z_{k_\ell}\|_q^2.$$

Averaging over all permutations yields

$$\left\| \sum_{j=0}^{r-1} Z_j \right\|_q^2 \leq \|Z_0\|_q^2 + \eta_r(q) \sum_{j=1}^{r-1} \|Z_j\|_q^2.$$

Finally,

$$\left\| \sum_j Z_j \right\|_q^q = \sum_{x=0}^{r-1} \left\| \sum_{j=0}^{r-1} \omega_r^{jx} A_j \right\|_q^q, \quad \|Z_j\|_q^2 = r^{2/q} \|A_j\|_q^2.$$

Dividing by $r^{2/q}$ gives the displayed inequality. The finite direct-sum case follows by the same block-diagonal embedding. $\square$

Main Theorem

Theorem 1 (Physical-space matrix hypercontractivity for $q \geq 2$). *Let $r \geq 2$, $n, m \geq 1$, and $1 \leq p \leq 2 \leq q < \infty$. If*

$$0 \leq \rho \leq \left(\frac{\zeta_r(p)}{\eta_r(q)} \right)^{1/2},$$

then every $f : \mathbb{Z}_r^n \rightarrow \mathbb{C}^{m \times m}$ satisfies

$$\|T_\rho f\|_{L_q(S_q)} \leq \|f\|_{L_p(S_p)}.$$

For $r = 2$, this is the classical threshold $\rho \leq \sqrt{(p - 1)/(q - 1)}$.

Proof. First consider one coordinate, so $f : \mathbb{Z}_r \rightarrow \mathcal{A}$ where $\mathcal{A}$ is a finite direct sum of matrix algebras. Write $f(x) = \sum_{k=0}^{r-1} \omega_r^{kx} A_k$. By Lemma 2,

$$\|T_\rho f\|_{L_q(S_q)}^2 \leq \|A_0\|_q^2 + \eta_r(q) \rho^2 \sum_{k=1}^{r-1} \|A_k\|_q^2.$$

Since $q \geq p$, Schatten monotonicity gives $\|A_k\|_q \leq \|A_k\|_p$. The assumed bound on ρ therefore implies

$$\|T_\rho f\|_{L_q(S_q)}^2 \leq \|A_0\|_p^2 + \zeta_r(p) \sum_{k=1}^{r-1} \|A_k\|_p^2.$$

Lemma 1 bounds the last expression by $\|f\|_{L_p(S_p)}^2$. This proves the one-coordinate estimate with values in any finite direct sum of matrix algebras.

For $G = \mathbb{Z}_r^n$, write $T_\rho = T_\rho^{(1)} \cdots T_\rho^{(n)}$, where $T_\rho^{(j)}$ acts on the j th coordinate. The single-coordinate noise kernel is the probability measure assigning mass $(1 + (r-1)\rho)/r$ to 0 and $(1-\rho)/r$ to each nonzero element, so $T_\rho^{(j)}$ is an $L_p(S_p)$ contraction for $0 \leq \rho \leq 1$.

Apply the one-coordinate estimate to the last coordinate, viewing the first $n-1$ coordinates and the matrix variable as one finite direct sum of matrix algebras. Since the remaining coordinate noises commute with the last one and are $L_p(S_p)$ contractions,

$$\begin{aligned} \|T_\rho f\|_{L_q(S_q)} &= \left\| T_\rho^{(n)} T_\rho^{(1)} \cdots T_\rho^{(n-1)} f \right\|_{L_q(S_q)} \\ &\leq \left\| T_\rho^{(1)} \cdots T_\rho^{(n-1)} f \right\|_{L_p(S_p)} \leq \|f\|_{L_p(S_p)}. \end{aligned}$$

This proves the theorem. □

Scope and Remaining Range

The theorem is a physical-space matrix-valued hypercontractivity statement over $\mathbb{Z}_r^n$ for every $1 \leq p \leq 2 \leq q < \infty$. It therefore answers the large-target-exponent part of the source question with an explicit dimension-free constant. It also recovers the optimal scalar threshold in the Boolean case $r = 2$ for this range.

The theorem does not settle $p < q < 2$. The simple smoothness/convexity sandwich used above is intrinsically pinned at the Hilbert exponent 2. Extending it below 2 appears to require sharper Schatten Hanner-type information; this is exactly the sort of terrain the source paper flags when discussing why matrix Hanner inequalities are only known in restricted ranges. Small random 2×2 tests did not reveal a counterexample, so this packet records the result as a partial theorem rather than a disproof of the remaining range.

References

1. S. Arunachalam and J. F. Doriguello, *Matrix hypercontractivity, streaming algorithms and LDCs: the large alphabet case*, arXiv:2109.02600; ACM Transactions on Computation Theory 16(4), Article 21, 2024.
2. K. Ball, E. A. Carlen, and E. H. Lieb, *Sharp uniform convexity and smoothness inequalities for trace norms*, Inventiones Mathematicae 115 (1994), 463–482.